UG (NEP) 6th SEMESTER
COMPUTER APPLICATIONS – COMPUTER ORGANIZATION & ARCHITECTURE
(CAP622J2)

Session: June 2026 No. of Questions: 09
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 36

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. Define instruction code.
2. What is data bus?
3. Define RISC.
4. What is microprogrammed control?
5. What is RAM?
6. What is direct mapping?
7. What is multicore architecture?
8. What is bus arbitration?

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Describe Von Neumann computer architecture with a suitable diagram.

OR

Describe the instruction cycle with a suitable example.

3. Discuss direct, indirect, register, and indexed addressing modes.

OR

Explain hardwired and microprogrammed control units.

4. Describe magnetic disks and magnetic tapes.

OR

Explain memory hierarchy with a suitable diagram.

5. Explain isolated I/O and memory-mapped I/O.

OR

Discuss parallel processing, multiprocessing systems, and multicore architecture.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Discuss bus interconnection structures and the common bus system in detail.
7. Compare RISC and CISC architectures. Explain instruction pipelining.
8. Discuss cache memory organization and different cache mapping techniques.
9. Discuss programmed I/O, interrupt-initiated I/O, and DMA techniques.